Regression and Residuals

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach each residual as a matched-unit comparison.

Supply all plotting points,explain code assumptions and distinguish observed patterns from fitted models or causal claims.

Keep one habit: Require subtraction evidence,units,sign interpretation and bounded summaries.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Distinguish numeric **prediction** from classification

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Teach two 45-minute sessions. This week focuses on **regression** and **residual**s. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

A category; **regression** predicts a numeric target instead.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Optional Python 3 editor; no external packages or AI accounts required

Opening retrieval

Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

Look for: Categorize development errors using **evidence**

Reference example, not the only acceptable wording: Teacher explanation: Dim 4/12 about 33.3 percent; bright 3/18 about 16.7 percent.

Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

Look for: Explain why overall accuracy can hide failure on a less common class

Reference example, not the only acceptable wording: The **training**-majority rule always predicts clear. Matrix [[16,0],[4,0]] gives 16/20=80 percent, FP=0,FN=4.

It misses all four blocked cases, so the overall score hides complete failure on that evaluated class.

Without opening the old lesson, explain one key idea from Train a Classifier. Give an example and a mistake that the idea helps you avoid.

Look for: Connect **training**-only scaling and neighbour **prediction** in one pipeline

Reference example, not the only acceptable wording: fit learns **training** minima and ranges and stores scaled **training** rows with IDs and labels. predict takes a raw feature pair and the fitted state.

Evaluation compares a **prediction** with its reference label afterward. Validation references belong only in evaluation, not in **fitting** or the neighbour pool.

End goal

By the end, I can explain **regression** and **residual**s using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Error Analysis

Accuracy and Imbalance

Train a Classifier

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Actual leaf length is 7 cm and **prediction** is 10 cm. Using **residual**=actual–predicted, calculate **residual** and absolute error. Interpret the sign.

Concept to explain

Regression predicts a number; a **residual** records signed error and reveals direction as well as size.

Term	Meaning
regression	prediction of a numerical target
prediction	model-estimated value
residual	actual value minus predicted value
residual plot	errors displayed against predictions or inputs

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach matched quantities and subtraction order first

> Keep numerical regression distinct from a software regression.

Target, source and sign convention

A school garden example predicts plant height in cm from recorded conditions. The target is numeric, so a **prediction** can be too high or too low by different amounts. Define **residual** as observed minus predicted.

Keep units and the sign convention visible; some software uses the opposite error convention.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **regression** and **residual**s.

Here regression means prediction of a numerical quantity. In last week's software-change record, a regression meant a new or worsened failure; those uses of the word are different. These height examples use supplied predictions, not a newly trained model. Observe and predict the same case, quantity, time and unit before subtracting. A day-7 height is not a growth increment without an initial-height observation.

Trace main cancellation example

Observed heights [4, 7, 9] cm and **prediction**s [5, 6, 9] cm give **residual**s [-1, 1, 0] cm. The first **prediction** is too high, the second too low and the third exact. Mean signed **residual** is zero, yet two **prediction**s are wrong.

Opposite errors can cancel.

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

Term	Meaning
Regression	Prediction of a numeric target
Residual	Observed target minus predicted target in this course
Underprediction	A prediction below the observed value
Overprediction	A prediction above the observed value

Regression and residuals at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Observed 4 predicted 5	Residual minus 1	Overprediction
Observed 7 predicted 6	Residual plus 1	Underprediction
Observed 9 predicted 9	Residual zero	Exact on this record

To inspect a residual plot, place a feature or prediction on the horizontal axis and residual on the vertical axis. A trend can suggest a missed relationship. It does not by itself identify the cause or validate a new model. Use development data for analysis and preserve an independent final test.

Why does mean residual zero not imply perfect predictions? Explain the steps or evidence behind your response.

Read separate pictured pairs and signs

Use **residual** signs to say whether each plant-height estimate is too high or low.

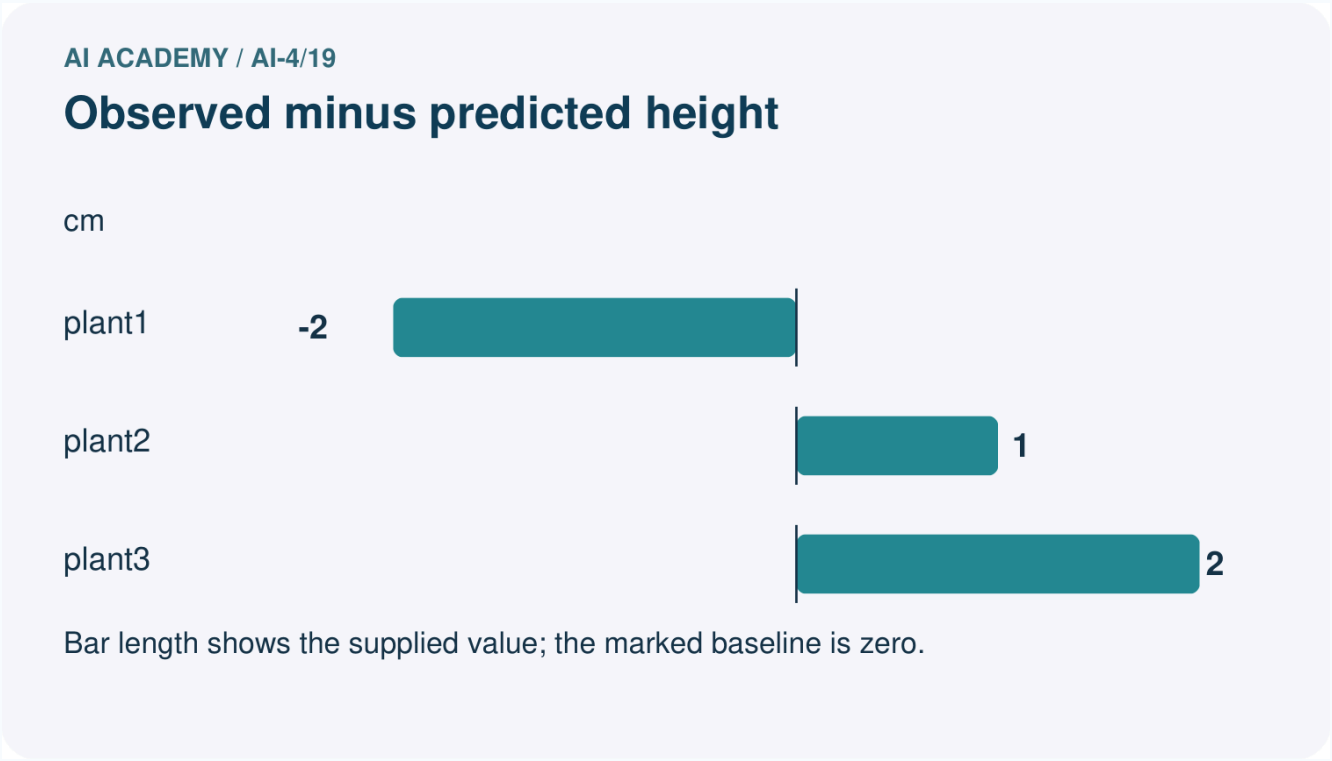
Residual=observed-predicted. Observed heights [5, 8, 12] cm; predicted [7, 7, 10] cm.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main observed4,7,9versus predicted5,6,9 gives residuals-1,+1,0cm,mean0and MAE2/3cm. Picture observed5,8,12versus predicted7,7,10 gives-2,+1,+2cm,mean1/3and MAE5/3cm. Raising only the pictured predictions by1cm gives-3,0,+1,mean-2/3and MAE4/3cm. A smaller

MAE on these already inspected records is a development observation,not proof of better future performance.

Keep observed values and the unit unchanged during this arithmetic comparison.



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace bars and raising predictions

AI ACADEMY / AI-4/19 · 02 · TRACE THE CASE

Follow the deciding steps

01 Subtract matched pairs

Residuals are5-7=-2, 8-7=1, 12-10=2 cm.

02 Read directions

First prediction is too high by2; second too low by1; third too low by2.

03 Avoid cancellation

Mean residual is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Subtract matched pairs	Residuals are5-7=-2, 8-7=1, 12-10=2 cm.
2. Read directions	First prediction is too high by2; second too low by1; third too low by2.
3. Avoid cancellation	Mean residual is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Always declare the **residual** sign convention; some software uses predicted minus observed.

Change: Raise all **prediction**s by1 cm.

Prediction: **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm.

Reason: Each **residual** decreases by1 under the observed-minus-predicted convention.

▼ Additional12observed/9predicted example

Actual height=12 cm, **prediction**=9 cm. **Residual** means actual minus predicted.

Calculate and interpret.

+3 cm means **underprediction** by 3 cm. State this sign convention consistently.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Plot **prediction**s and calculate positive and negative **residual**s.

MODEL IDEA **Regression** predicts a number; a **residual** records signed error and reveals direction as well as size.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Actual plant height is 30 cm; **prediction** is 27 cm. **Residual** = actual – predicted = 30 – 27 = +3 cm.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 Actual plant height is 30 cm; **prediction** is 27 cm.

STEP 2 **Residual** = actual – predicted = 30 – 27 = +3 cm.

STEP 3 Another case has actual 20 and predicted 24, **residual** –4 cm.

STEP 4 Plot **residual**s; a pattern above or below zero suggests systematic model error.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Exact explanation to model

Explain the added worked example

Key terms

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Regression and Residuals

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide the calculation, then remove help

> Collect each learner's prediction before showing its key.

Current completion task and checked key

Completion task: Why does mean **residual** zero not imply perfect **prediction**s?

Teacher check: Teacher explanation: Negative and positive **residual**s cancel even when individual errors are nonzero.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Regression** and **residual**s. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: **Residual**s -1, 1, 0 cm; the first record is overpredicted.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Which statistic reveals the size of errors without cancellation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

MAE: original5/3 cm and changed4/3 cm. Signed mean reports average direction and can hide offsetting errors. Changed-case check: **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm.

Each **residual** decreases by1 under the observed-minus-predicted convention.

Changed-case reasoning: **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm. Each **residual** decreases by1 under the observed-minus-predicted convention.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: Actual height=12 cm, **prediction**=9 cm. **Residual** means actual minus predicted.

Write the question to explain: Calculate and interpret.

Reveal after **prediction**s: +3 cm means **underprediction** by 3 cm. State this sign convention consistently.

Use the earlier transfer question as additional practice. Reserve the selected fresh three-pair signed-**residual** task for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided response: Negative and positive **residual**s cancel even when individual errors are nonzero. Exit response: Convert to a shared unit; both represent 2 cm. Document units before comparing or aggregating errors.

Case-specific teaching sequence

Stage · Teacher action

> **Stage:** Session 1 start to 7 minutes
>
> **Teacher action:** Use the prerequisite question: What does a class label predict? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify regression in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: calculate residuals; explain cancellation. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Trace the Python check after the paper calculation.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between regression and residual. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Interpret residuals -2, 0, 3 ordered by growth stage.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Distinguish numeric prediction from classification; Calculate signed residuals with a stated convention. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Six-case lab mean0with all6predictions wrong,MAE14/6=7/3cm. Plot residual versus predicted height;constructed larger magnitudes suggest a check,not a proved general pattern. Missing observed outcome has no residual;separate32vs12cm extreme gives+20and needs source checking. Original zip code assumes equal-length matched nonempty lists.

Selected observed6,10,13/predicted8,9,11 gives-2,+1,+2cm,mean1/3;first overpredicts,others underpredict. No model is fitted.

Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. **Residual** Walk: calculate signed errors, place them on a plot, and identify bias, outliers, or changing spread.
3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.
5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: **Residual**s -1, 1, 0 cm; the first record is overpredicted.

A stronger claim makes a stronger project: Convert to a shared unit; both represent 2 cm. Document units before comparing or aggregating errors.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: **Residual**s [-2, 1, 2] cm; mean 1/3 cm. First overpredicted, others underpredicted.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the selected fresh three-pair signed-residual task for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR Positive and negative **residual**s cancel, so their average proves accuracy.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

Selected independent task and checked key

Selected case: Observed [6, 10, 13], predicted [8, 9, 11] cm. Give **residual**s, mean **residual** and error directions.

Teacher explanation: **Residual**s [-2, 1, 2] cm; mean 1/3 cm. First overpredicted, others underpredicted.

Judge trace, scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Calculate residuals

Compute the worked **residual**s and identify **overprediction**.

Teacher explanation: **Residual**s -1, 1, 0 cm; the first record is overpredicted.

W2 Explain cancellation

Why does mean **residual** zero not imply perfect **prediction**s?

Teacher explanation: Negative and positive **residual**s cancel even when individual errors are nonzero.

W3 Read a pattern

Interpret **residual**s -2, 0, 3 ordered by growth stage.

Teacher explanation: They shift from **overprediction** to **underprediction**; investigate a missed relationship without claiming a proven cause.

W5 Check units

A report combines 2 cm and .02 m as if they were different magnitudes. Repair it.

Teacher explanation: Convert to a shared unit; both represent 2 cm. Document units before comparing or aggregating errors.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

+3 cm means **underprediction** by 3 cm. State this sign convention consistently.

Additional transfer practice question

Define **residual** as actual minus predicted. Actual height is 14 cm and **prediction** is 17 cm. Calculate **residual** and interpret its sign.

Reasoned practice answer

Residual is 14–17 = −3 cm. The model overpredicts by 3 cm under the stated convention.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A negative **residual** means a negative physical height.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Residual=-3 cm means **overprediction** by 3 cm. Absolute error=3 cm. State the sign convention because some sources reverse it; keep units and convention consistent across plots and calculations.

Explanations for the additional workbook examples

W1 Calculate residuals

W2 Explain cancellation

W3 Read a pattern

W4 Apply independently

Observed [6, 10, 13], predicted [8, 9, 11] cm. Give **residual**s, mean **residual** and error directions.

Teacher explanation: **Residual**s [-2, 1, 2] cm; mean 1/3 cm. First overpredicted, others underpredicted.

W5 Check units

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Use a miniature growth-**prediction** side experiment to distinguish **regression evidence** from the main classifier metrics.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **regression** and **residual** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: **Residual**s [-2, 1, 2] cm; mean 1/3 cm. First overpredicted, others underpredicted. Fresh retry observed [3, 8, 12], predicted [4, 8, 10]: **residual**s [-1, 0, 2], mean 1/3 in the target unit.

Require sign interpretations. Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Delayed return

Suggested return points in this level: ai-4/20, ai-4/22, ai-4/25. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	

Record	Teacher evidence
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 20: Mean, Median, and Baselines. Retrieve today's **evidence** and identify what carries forward and what is new.

Next week: Constant **prediction** baselines. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Distinguish numeric **prediction** from classification

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Why does mean **residual** zero not imply perfect **prediction**s?

Reasoned practice answer: Teacher explanation: Negative and positive **residual**s cancel even when individual errors are nonzero.

Check every learner before discussion: Use the worked case for **Regression** and **residual**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: **Residual**s -1, 1, 0 cm; the first record is overpredicted.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Raise all **prediction**s by1 cm. Require a **prediction** and a reason before comparing. Expected result: **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm.

Reason: Each **residual** decreases by1 under the observed-minus-predicted convention.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Earlier wording, alternatives and picture transcripts**

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

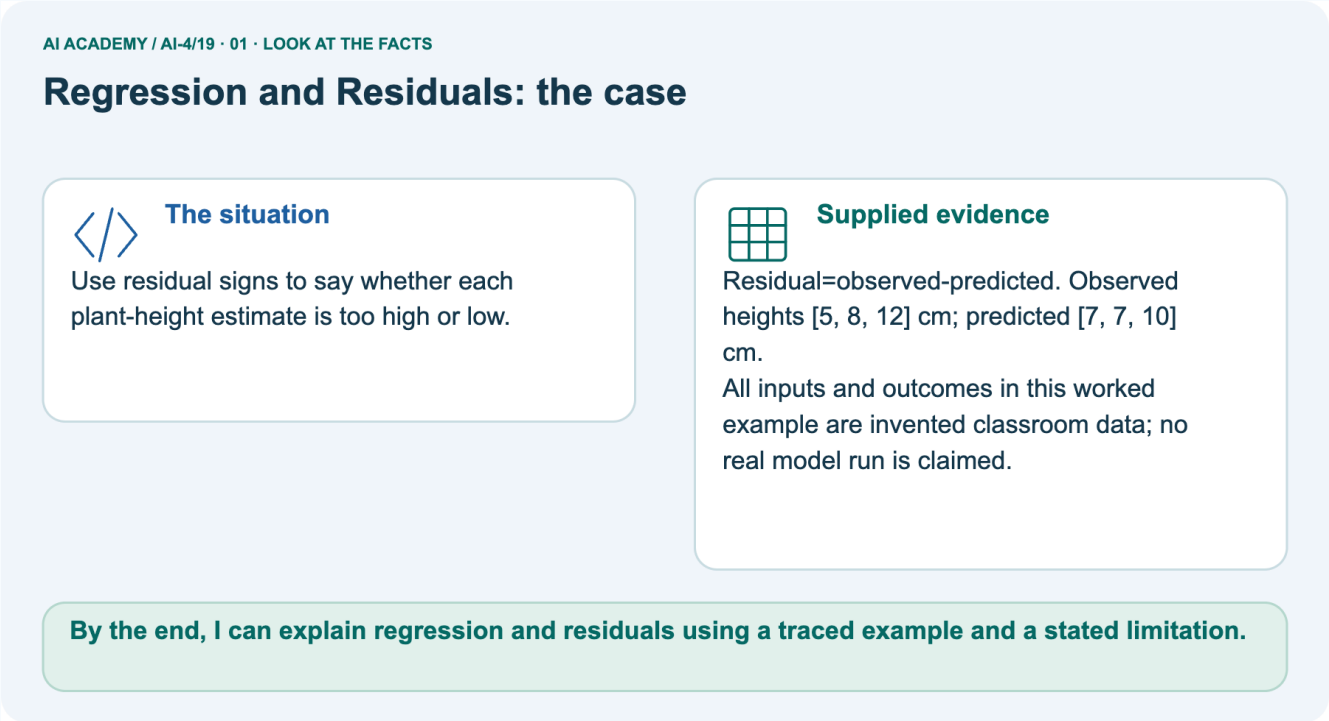
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Regression and Residuals: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

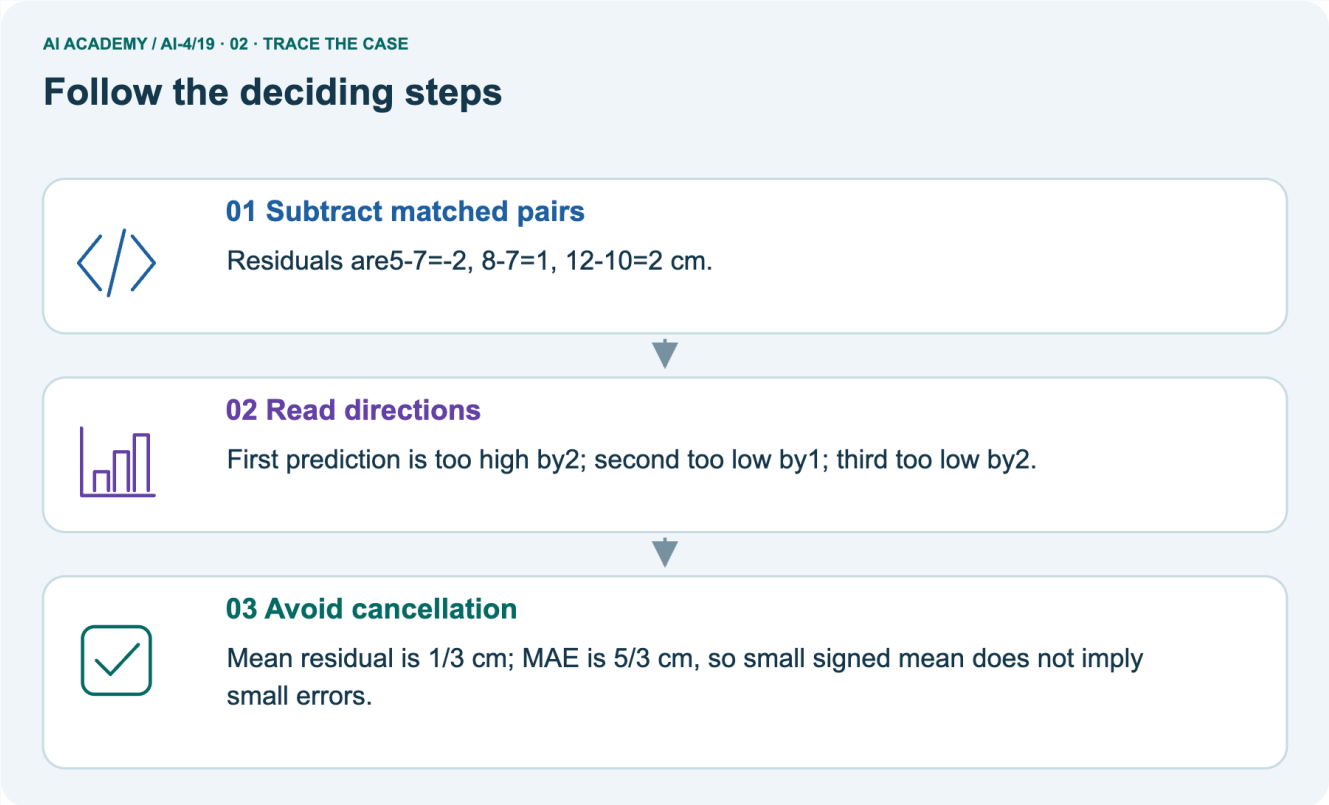
Use **residual** signs to say whether each plant-height estimate is too high or low. Supplied facts: **Residual**=observed-predicted. Observed heights [5, 8, 12] cm; predicted [7, 7, 10] cm.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



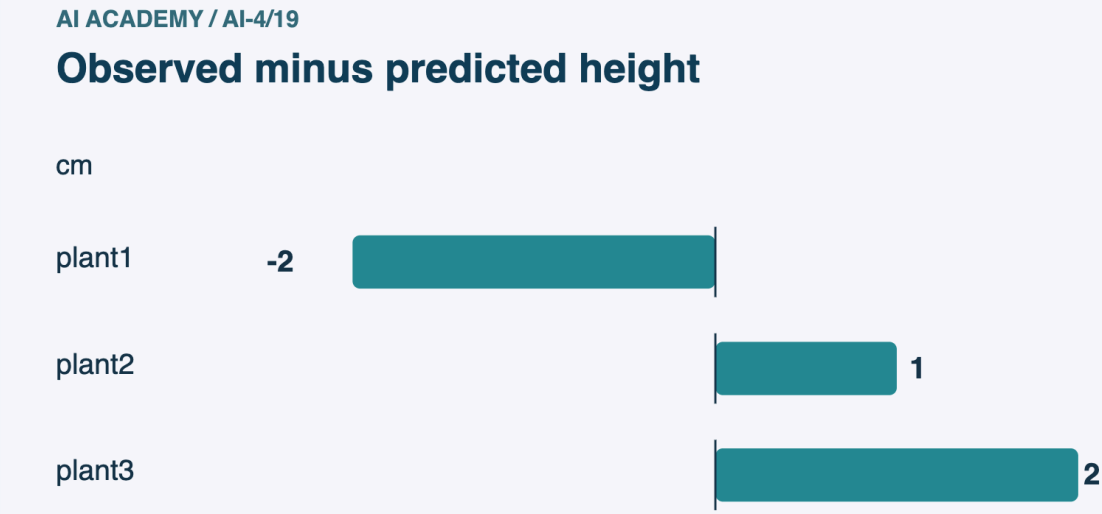
Follow the labelled picture with the matching explanation. Open the vector diagram

Subtract matched pairs: **Residual**s are5-7=-2, 8-7=1, 12-10=2 cm. → Read directions: First **prediction** is too high by2; second too low by1; third too low by2. → Avoid cancellation: Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Observed minus predicted height



Follow the labelled picture with the matching explanation. Open the vector diagram

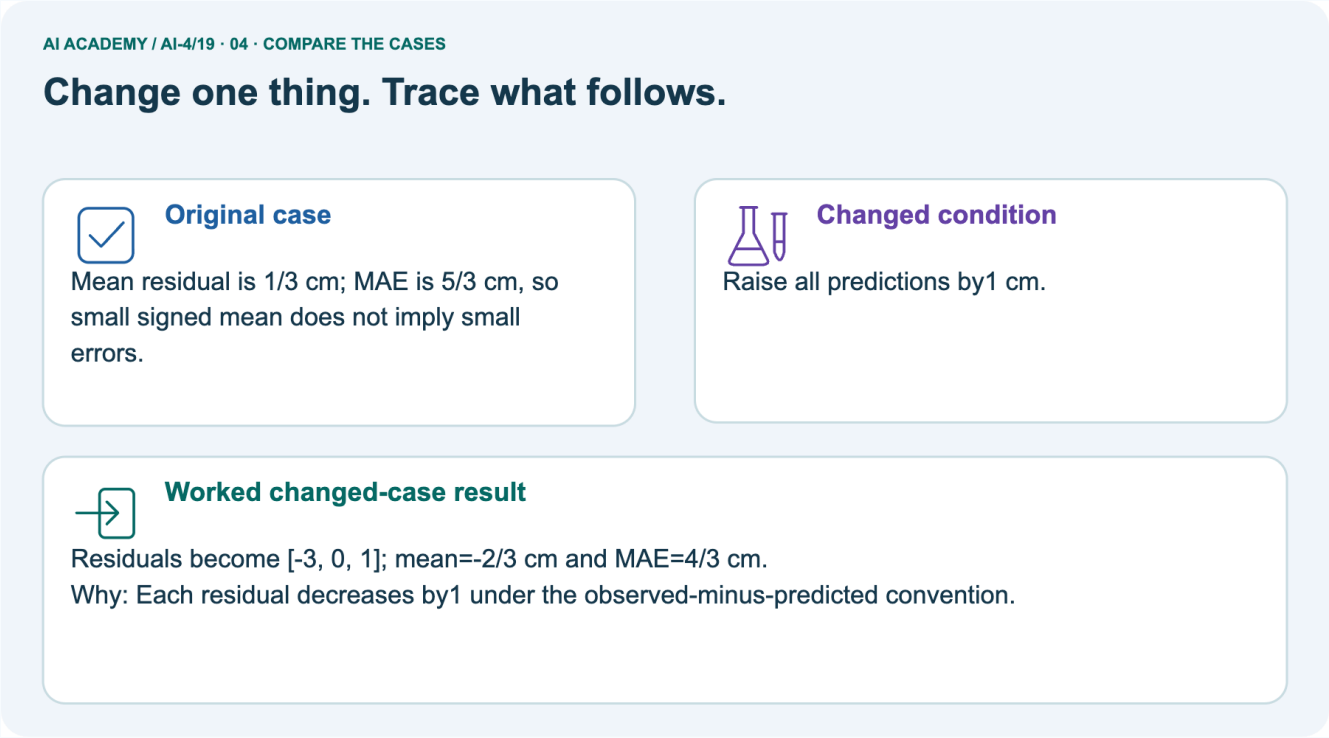
Observed minus predicted height. Units: cm. plant1: -2. plant2: 1. plant3: 2.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. Changed condition: Raise all **prediction**s by1 cm.

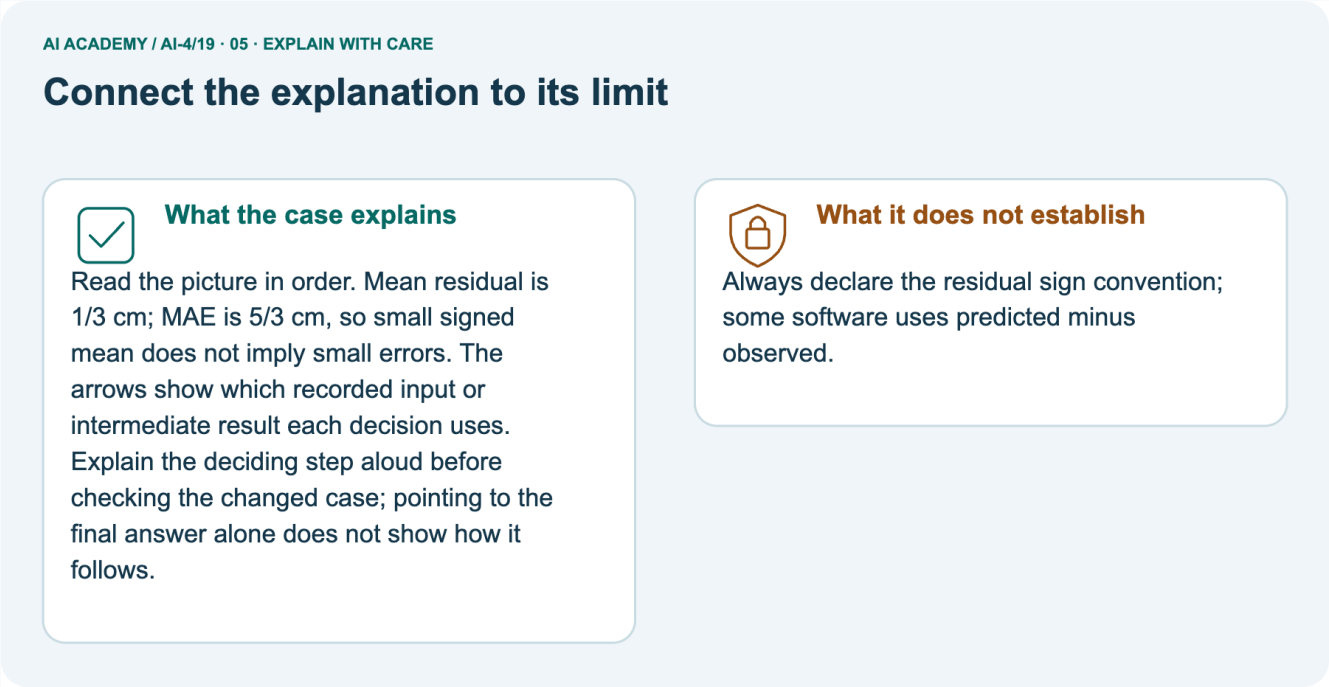
Expected result: **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm. Reason: Each **residual** decreases by1 under the observed-minus-predicted convention.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Each **residual** decreases by1 under the observed-minus-predicted convention.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

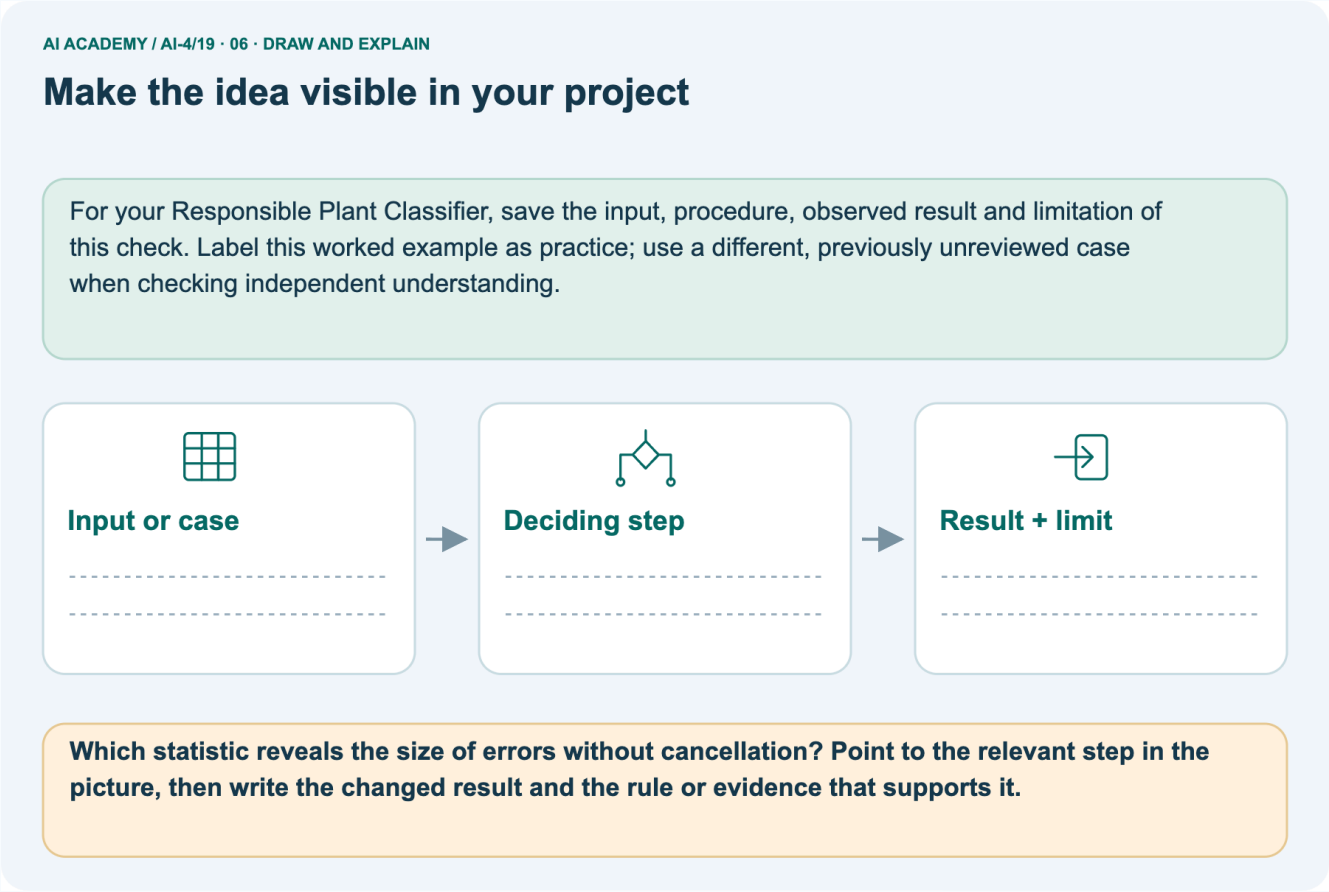
Limit: Always declare the **residual** sign convention; some software uses predicted minus observed.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Always declare the **residual** sign convention; some software uses predicted minus observed.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which statistic reveals the size of errors without cancellation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: MAE: original5/3 cm and changed4/3 cm. Signed mean reports average direction and can hide offsetting errors. Changed-case check: **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm.

Each **residual** decreases by1 under the observed-minus-predicted convention.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Regression and Residuals: the case

02. Follow the deciding steps

03. Observed minus predicted height

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.